





Timeline View: CollNet

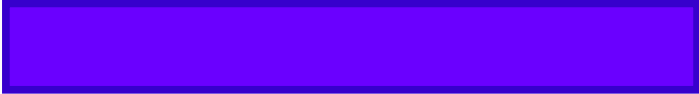
S0: 机内流 (NVLink/PCIe)

Local Reduce + Local Broadcast / Local Reduce-Scatter + Local All-gather



S1: 集群内流 (RDMA/InfiniBand)

Intra-domain All-Reduce



S2: 集群间流 (Wide Area Network)

Inter-domain All-Reduce (高往返时延 / 低带宽)



Timeline View: chunk c0-c3, All-Reduce and Broadcast



S0: 机内流 (NVLink/PCIe)

Local Reduce + Local Broadcast / Local Reduce-Scatter + Local All-gather



S1: 集群内流 (RDMA/InfiniBand)

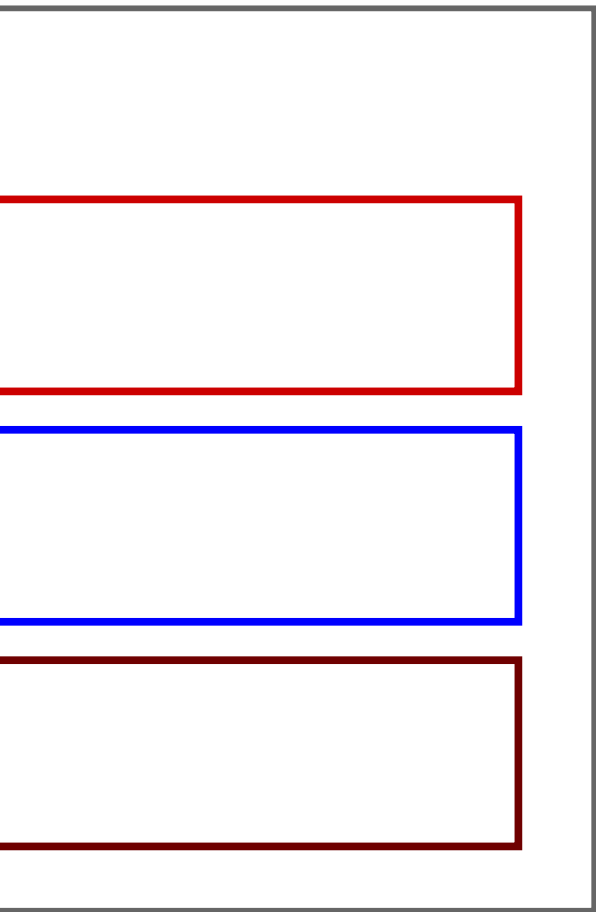
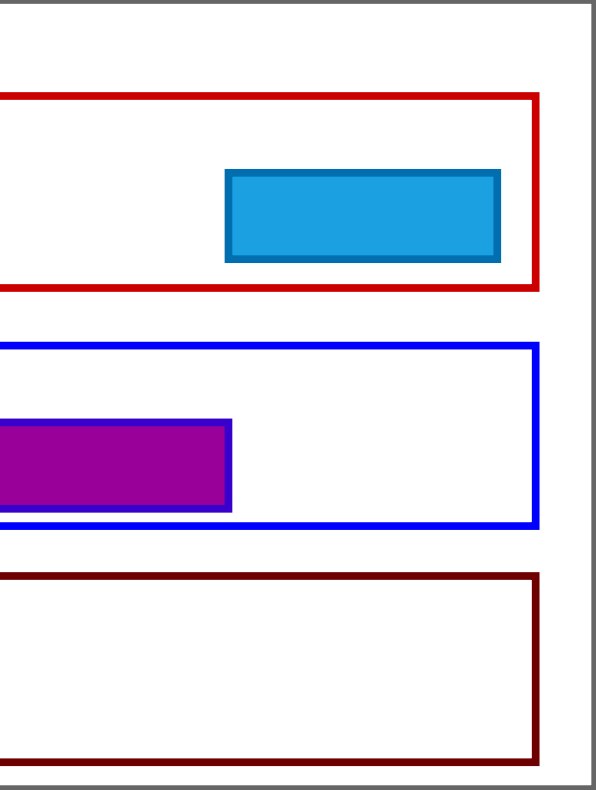
Intra-domain All-Reduce



S2: 集群间流 (Wide Area Network)

Inter-domain All-Reduce (高往返时延 / 低带宽)





Legend and Mapping

Local All-Reduce  
(域内聚合)

Local Broadcast  
(域内广播)

Intra-domain Reduce  
(集群内)

Intra-domain Reduce  
(集群内)

Inter-domain AR  
(集群间)